

Phishing Email Investigation with Malicious Attachment

Overview

This home lab was completed using a Linux virtual machine while working through a TryHackMe phishing investigation scenario.

The objective was to determine whether an email requesting a financial transfer was legitimate or part of a phishing campaign.

Environment

Platform: TryHackMe

Operating System: Linux Virtual Machine

Role: SOC Analyst (Level 1)

Tools Used

VirusTotal

MXToolbox

WHOIS Lookup

DMARCian

Linux SHA256

Email Header Analysis

Investigation Steps

Header Analysis

I examined the email headers to identify:

Sender address: info@mutawamarine.com

Return-Path: info@mutawamarine.com

Originating IP: **192.119.71.157**

Subject :The sender requested an unexpected money transfer, which immediately raised suspicion.

Recipient :webmaster@redacted.org

Reply-To email address: No

Timestamp:6/10/20, 05:58

IP Reputation

The originating IP address was investigated using:

MXToolbox

WHOIS

This helped determine the reputation and ownership of the sending server.

IP Details For: 192.119.71.157

Decimal: 3229042589

Hostname: client-192-119-71-157.hostwindsdns.com

ASN: 54290

ISP: HostPapa

Services: Data Center/Transit

Country: United States

State/Region: New York

City: Buffalo

Latitude: 42.8868 (42° 53′ 12.35″ N)

Longitude: -78.8787 (78° 52′ 43.24″ W)



CLICK TO CHECK BLACKLIST STATUS

LISTED	BARRACUDA	192.119.71.157 was listed	Detail	900	78	Ignore
LISTED	FABELSOURCES	192.119.71.157 was listed	Detail	600	31	Ignore
LISTED	RATS Spam	192.119.71.157 was listed	Detail	2100	0	Ignore
OK	0SPAM				78	
OK	0SPAM RBL				78	

Checking **192.119.71.157** against **60** known blacklists...
Listed **3** times with **0** timeouts



Whois IP 192.119.71.157

```
#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy\_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#
```

```
NetRange:      192.119.64.0 - 192.119.127.255
CIDR:          192.119.64.0/18
NetName:       HOSTWINDS-18-2
NetHandle:     NET-192-119-64-0-1
Parent:        NET192 (NET-192-0-0-0-0)
NetType:       Direct Allocation
OriginAS:
Organization:  HostPapa (HOSTP-7)
RegDate:       2012-11-12
Updated:       2026-05-13
Comment:       Geofeed https://geofeeds.oniaas.io/geofeeds.csv
Ref:           https://rdap.arin.net/registry/ip/192.119.64.0
```

```
OrgName:       HostPapa
OrgId:         HOSTP-7
Address:       325 Delaware Avenue
Address:       Suite 300
City:          Buffalo
StateProv:     NY
PostalCode:    14202
Country:       US
RegDate:       2016-06-06
Updated:       2025-10-05
Ref:           https://rdap.arin.net/registry/entity/HOSTP-7
```

SPF and DMARC Validation

I validated the sender's authentication records.

The investigation revealed:

SPF = Fail

DMARC = Unknown

These findings indicate that the sender failed email authentication checks and may have been spoofed.

SPF Surveyor

Our **SPF Surveyor** provides a graphical view of DNS records that allows you to more easily identify which IPs are authorized to send on behalf of a domain. The tool will also verify that the SPF record has correct that could lead to diminished email deliverability and failed email authentication.

SURVEY DOMAIN

v=spf1 redirect=_spf.mail.com				
Prefix	Type	Value	PrefixDesc	Description
	v	spf1		The SPF record version
+	redirect	_spf.mail.com	Pass	The SPF record for Value replaces the current record.

DMARC Domain Checker

DMARC Domain Checker to quickly find out if a domain is properly protected against phishing, spoofing and domain abuse. It inspects DMARC, SPF and DKIM records to see if any issues are present.

mutawamarine@mail.com

CHECK DOMAIN

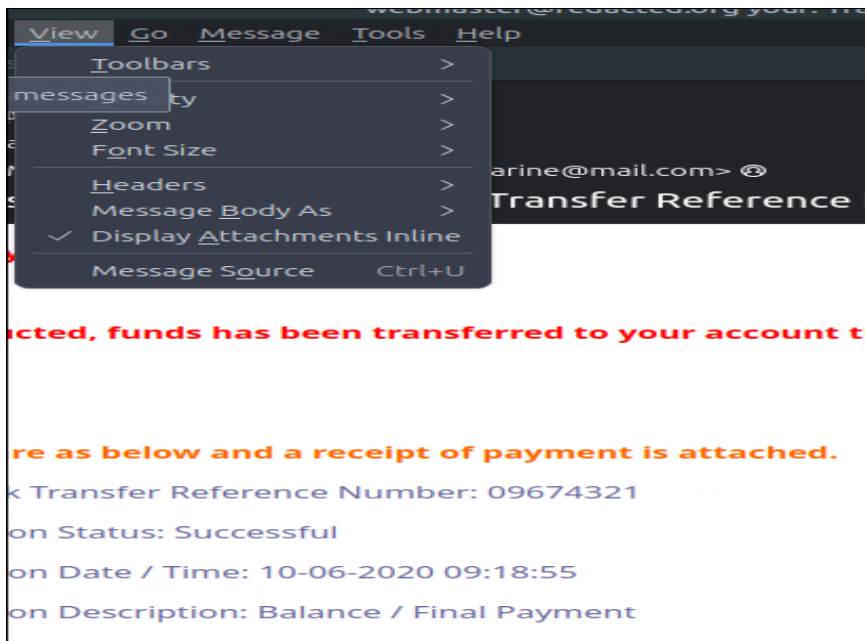
v=DMARC1; p=quarantine; rua=mailto:dmarcreport@mail.com

Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	quarantine	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.
rua	mailto:dmarcreport@mail.com	Receivers	Addresses to which aggregate feedback is to be sent. Comma separated plain-text list of DMARC URIs.

spf=fail smtp.mailfrom=mutawamarine.com;

dmARC=unknown

or we can check form message source :



```
Source of: file:///home/ubuntu/Desktop/challenge.eml? - Mozilla Thunderbird
File Edit View Help
X-Atlas-Received: from 10.201.192.162 by atlas125.free.mail.bf1.yahoo.com with http; Wed
Return-Path: <info@mutawamarine.com>
Received: from x.x.x.x (EHLO sub.redacted.com)
by atlas125.free.mail.bf1.yahoo.com with SMTPs; Wed, 10 Jun 2020 05:58:55 +0000
X-Originating-Ip: [x.x.x.x]
Received-SPF: fail (domain of mutawamarine.com does not designate x.x.x.x as permitted s
Authentication-Results: atlas125.free.mail.bf1.yahoo.com;
spf=fail smtp.mailfrom=mutawamarine.com;
dmarc=unknown
X-Apparently-To: redacted@yahoo.com; Wed, 10 Jun 2020 05:58:55 +0000
X-YMailISG: CA2X0WoWLDuMav_xVT1F_okXM35Y6SWpmP6zsE6LeQRxoxw4
YjzuEZUWxEEJzHhUGbKbpzCq7GFztoIFDbqKMKwunxnYA6aofbh6xusqm_FJ
x591PPWDY5NhhvW7H.Pwb9o9VmzNhbgKs3KzMN9I07Uh5jf5y6rUw.dSshjuv
j1RgxZYshquA.RCedSbTlM1pyxBT4LoSfMkWr0E4FgJSW3l9zg8wK35sizWP
GqFHAYID0v.G0U7dBURvMp8asqQiPa4kYC7v0oQTvmUEDtENPAjmCnfcajpo
gei5zs471gDrr3JWxiUMUTyChHRw9nCcZLepgGA2Jt_MdbCZ7qgFqWMvvo1I
nIXkl35mwKec90ZCIPJc6tCHAQyFkE.030_.0VmK_brmLt5oqQiGBYmyCV2i
CwhdwdTwYkUIIdgler50ESBs5mHXSqnNvtmpQoRjMPdqSXiB7yvSIFaiF5rQ8
0TEw0w1CkWz4gxhNU4FH4Iub03b9TLvUoX9KLEfx3Del5yPTF8xXY7NY_kzA
aCwKTjp4FaeT2Mk1Pq5P48DF.dB6hdMTmCoowuSwouW2M9Yp4euqKNzGrLcf
2KcRMR0fVcKdWXSaHw4tMhKvXSH0KiWMVfPXPamMt2c0cklVpwZMyql8w8W
PlponB1e66yiIqNuYV.vt64i52HFh0jNwcuuohMo7MA7DmMP.0tkdwLLUqLS
68Afikwfw3Sppf_pTqtP6NPf2wuAsJiaT7_QQ.4x3khgYrC4jTmXjVBWDVRV
wT0AdlQ716U8TVp.0AKvevKfMzfZotOTsGLuQU.w8uZhv_6mwKB4sW7Pbhbr
B1RJadC0va2CJiCbAC1Qnapm5egSIEqxkhboy8i0Ufz0qkD1a1_tn5nv1IXD
bCqer07cnAjpN1amfUvC8qjD345qb6k9l7h7a8TFsv_67Nkrok_.M4_MRZcf
```

Attachment Analysis

The attachment was downloaded inside a secure sandbox environment.

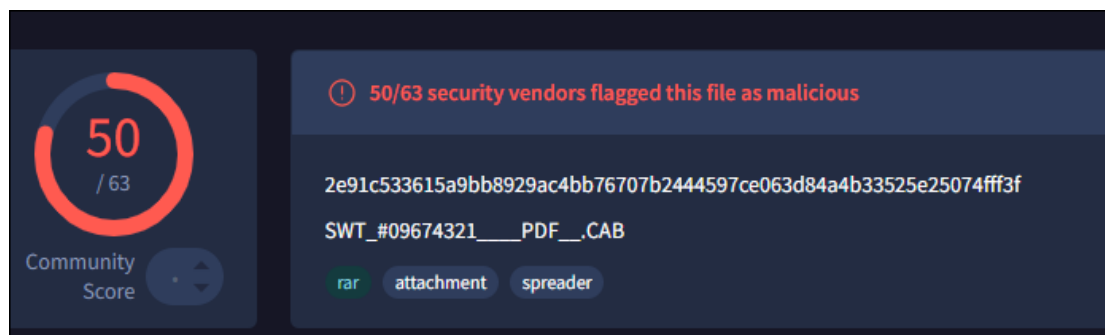
The filename was:

SWT_#09674321____PDF__

A SHA256 hash was generated using:

```
File Edit View Search Terminal Help
ubuntu@tryhackme:~/Desktop/SWT_#09674321____PDF__$ ls
SWT_#09674321__PDF.com
ubuntu@tryhackme:~/Desktop/SWT_#09674321____PDF__$ shasum -a 256 SWT_#09674321__
PDF.com
3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 SWT_#09674321__
PDF.com
ubuntu@tryhackme:~/Desktop/SWT_#09674321____PDF__$
```

[The hash was then submitted to VirusTotal to determine whether the attachment had previously been identified as malicious.](#)



MD5	f4dd3456cdb1976a145c1179a4d461ec
SHA-1	5a2bb8188377c15c036843b4a6ab9b0c0f2c1607
SHA-256	2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f
SSDEEP	12288:Mj6ygt8RoYqMAnuL8l0A81aBYolm9+X3B4k56:EgqRJCuL87toIC+X3O
TLSH	T12C94238893562439A8F7385DAFD0CFB5EFE898E74E8F97709CFD609E5D140446205AC
File type	RAR compressed rar
Magic	RAR archive data, v5
TrID	RAR compressed archive (v5.0) (61.5%) RAR compressed archive (gen) (38.4%)
Magika	RAR
File size	400.26 KB (409868 bytes)

Skills Demonstrated

Email Header Analysis

Malware Analysis

Attachment Investigation

IOC Collection

Threat Intelligence

Linux

VirusTotal

SPF / DMARC Analysis

Digital Forensics

Key Learning Outcomes

Learned how attackers spoof legitimate organizations.

Investigated sender authentication failures.

Verified IP reputation and domain ownership.

Analysed suspicious attachments safely within a virtual machine.

Used threat intelligence platforms to validate indicators of compromise.

Conclusion

This investigation followed the workflow of a SOC Level 1 analyst, beginning with email header analysis and progressing through IP reputation checks, authentication validation, and attachment analysis. Using multiple security tools improved confidence in identifying phishing indicators and reinforced practical incident investigation skills.